STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-EXTENSION-BY-ZERO-GRADED

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1088-1097 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 1099-1157, the complete author proof (57 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 13 of queue_sdga.json (pb=2029, stmt 1088-1097, proof 1099-1157).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0E8I: sources/stacks-sdga/pinned-original-source.tex, source lines 1157–1157. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03DH: sources/stacks-sdga/pinned-original-source.tex, source lines 1157–1157. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In the situation above we have*

$$\mathrm{Hom}_{Mod^{gr}(\mathcal{A})}(j_!\mathcal{M}, \mathcal{N}) = \mathrm{Hom}_{Mod^{gr}(\mathcal{A}_U)}(\mathcal{M}, j^*\mathcal{N})$$

Proof. By the discussion in Modules on Sites, Section 03DH the functors $j_!$ and j^* on $\mathcal{O}$-modules are adjoint. Thus if we only look at the $\mathcal{O}$-module structures we know that

$$\mathrm{Hom}_{\mathrm{Gr}^{gr}(Mod(\mathcal{O}))}(j_!\mathcal{M}, \mathcal{N}) = \mathrm{Hom}_{\mathrm{Gr}^{gr}(Mod(\mathcal{O}_U))}(\mathcal{M}, j^*\mathcal{N})$$

(Recall that $\mathrm{Gr}^{gr}(Mod(\mathcal{O}))$ denotes the graded category of graded $\mathcal{O}$-modules.) Then one has to check that these identifications map the $\mathcal{A}$-module maps on the left hand side to the $\mathcal{A}_U$-module maps on the right hand side. To check this, given

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$\mathcal{O}_U$-linear maps $f^n : \mathcal{M}^n \to j^*\mathcal{N}^{n+d}$ corresponding to $\mathcal{O}$-linear maps $g^n : j_!\mathcal{M}^n \to \mathcal{N}^{n+d}$ it suffices to show that

$$\begin{array}{ccc} \mathcal{M}^n \otimes_{\mathcal{O}_U} \mathcal{A}_U^m & \xrightarrow[f^n \otimes 1]{} & j^*\mathcal{N}^{n+d} \otimes_{\mathcal{O}_U} \mathcal{A}_U^m \\ \downarrow & & \downarrow \\ \mathcal{M}^{n+m} & \xrightarrow{f^{n+m}} & j^*\mathcal{N}^{n+m+d} \end{array}$$

commutes if and only if

$$\begin{array}{ccc} j_!\mathcal{M}^n \otimes_{\mathcal{O}} \mathcal{A}^m & \xrightarrow[g^n \otimes 1]{} & \mathcal{N}^{n+d} \otimes_{\mathcal{O}} \mathcal{A}_U^m \\ \downarrow & & \downarrow \\ j_!\mathcal{M}^{n+m} & \xrightarrow{g^{n+m}} & \mathcal{N}^{n+m+d} \end{array}$$

commutes. However, we know that

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{M}^n \otimes_{\mathcal{O}_U} \mathcal{A}_U^m, j^*\mathcal{N}^{n+d+m}) &= \mathrm{Hom}_{\mathcal{O}}(j_!(\mathcal{M}^n \otimes_{\mathcal{O}_U} \mathcal{A}_U^m), \mathcal{N}^{n+d+m}) \\ &= \mathrm{Hom}_{\mathcal{O}}(j_!\mathcal{M}^n \otimes_{\mathcal{O}} \mathcal{A}^m, \mathcal{N}^{n+d+m}) \end{aligned}$$

by the already used Modules on Sites, Lemma 0E8I. We omit the verification that shows that the obstruction to the commutativity of the first diagram in the first group maps to the obstruction to the commutativity of the second diagram in the last group. $\square$